

Dual-Foundation Models for Unsupervised Domain Adaptation

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01 Unsupervised Domain Adaptation

• **Problem:** A model trained on labeled synthetic data often performs poorly on unlabeled real-world images due to the domain gap.



Source: labeled synthetic → Target: unlabeled real

Two limitations of prior UDA methods

Pixel level Mixing-based self-training keeps only high-confidence pseudo-labels, so the student sees a small subset of target pixels.

Feature level Prototypes from a source-trained teacher are source-biased, unstable, and collapse for rare classes.

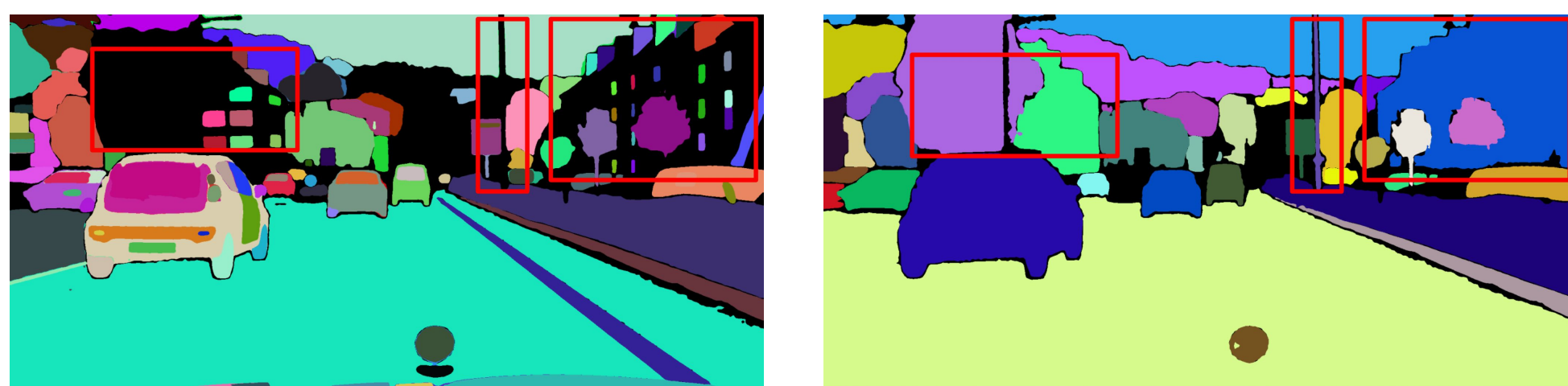


02 Key Insight

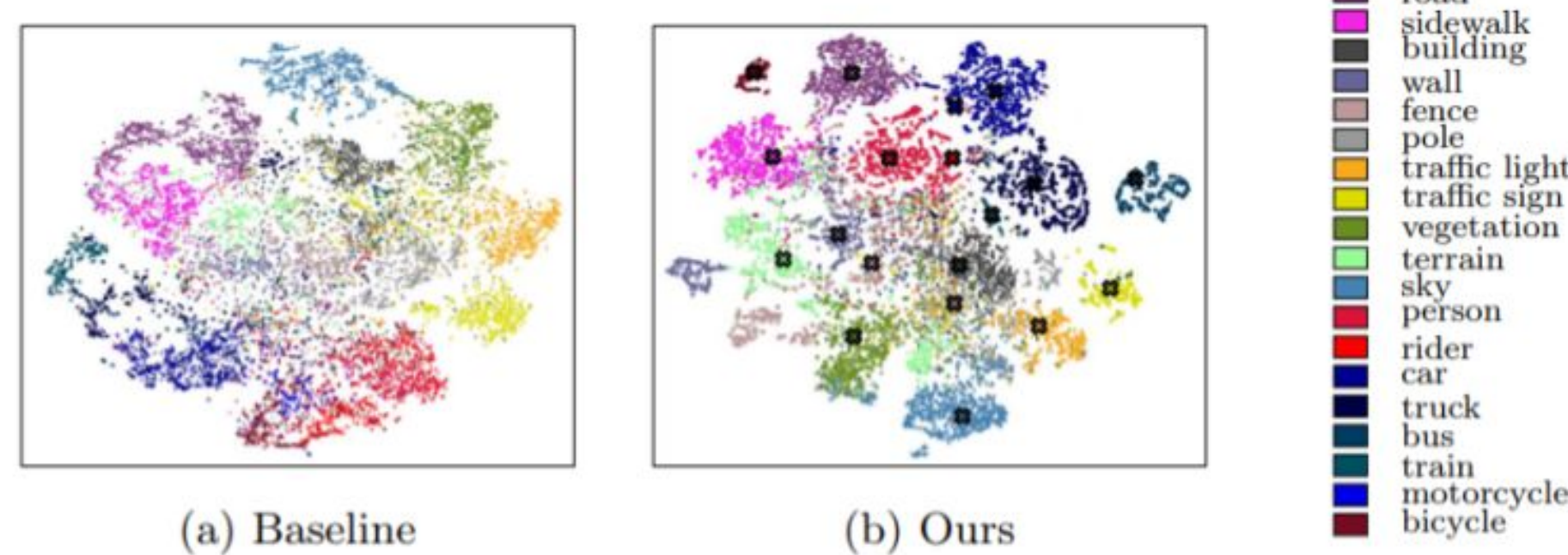
Super Pixel — Effective Prompts for SAM



SAM — object-level spatial priors



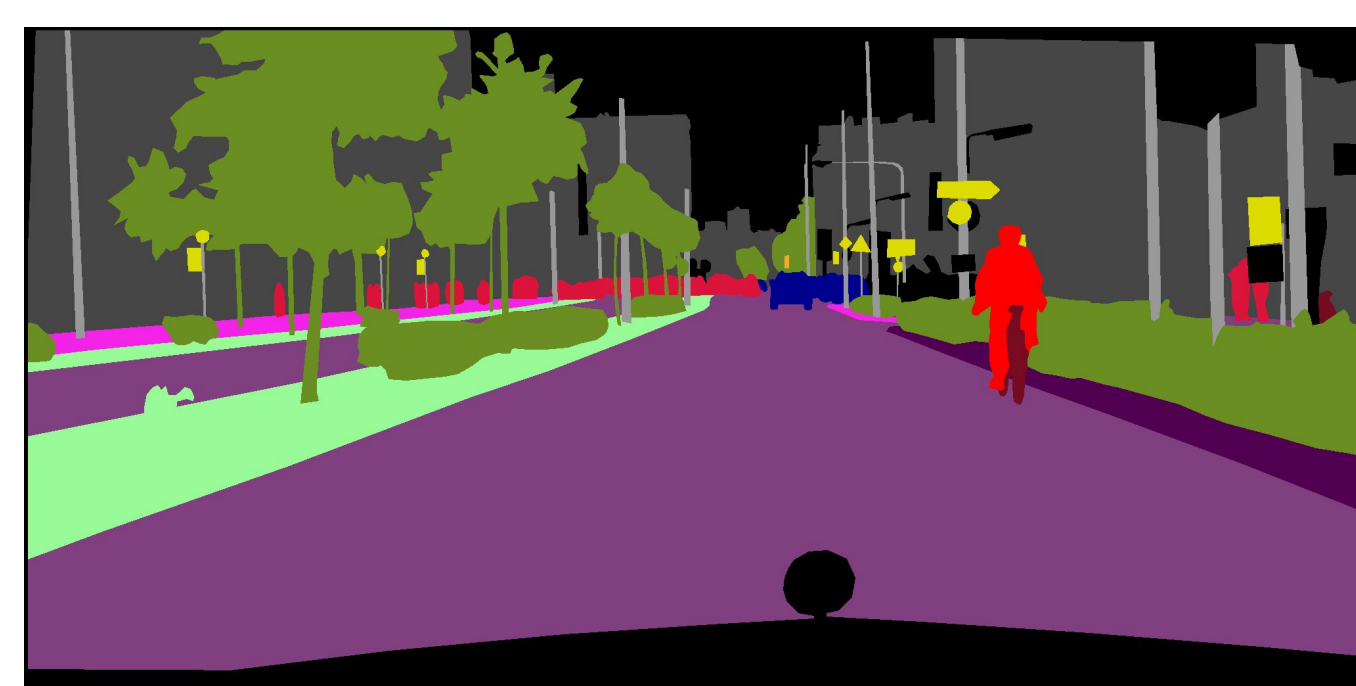
DINOv3 — domain-robust features



06 Qualitative Results



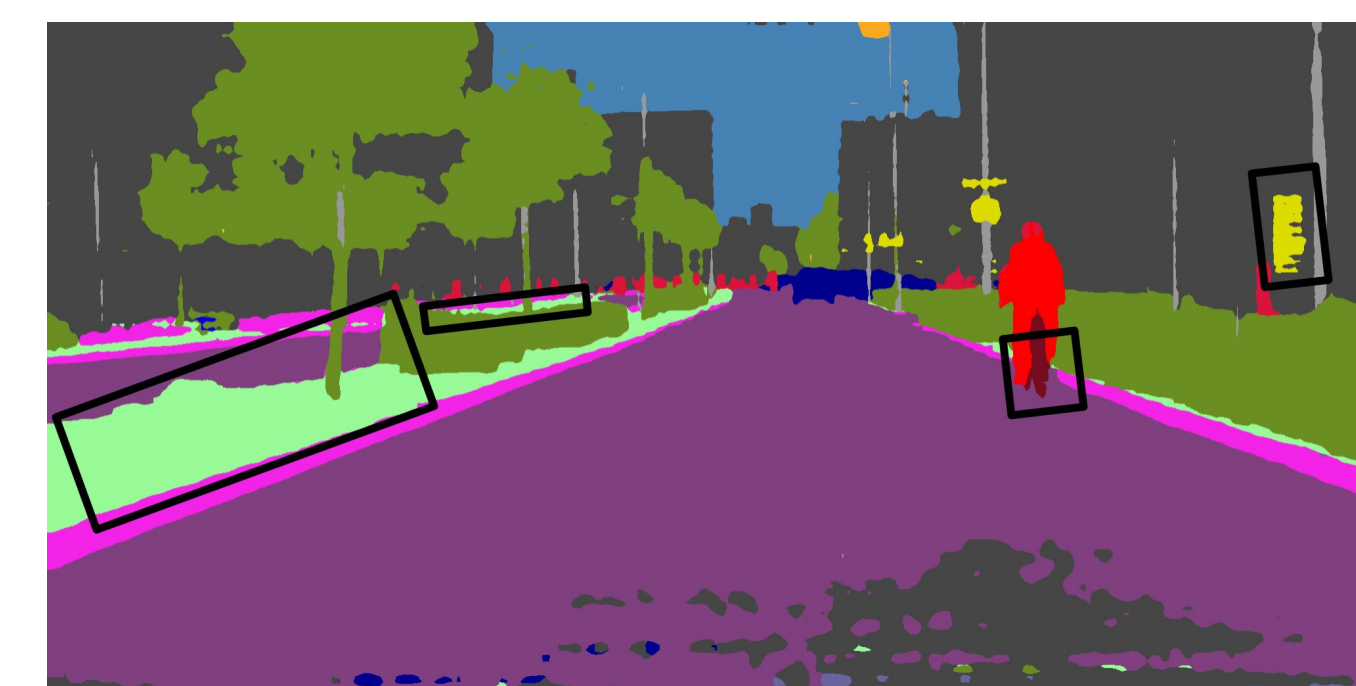
Target Image



Target Ground Truth

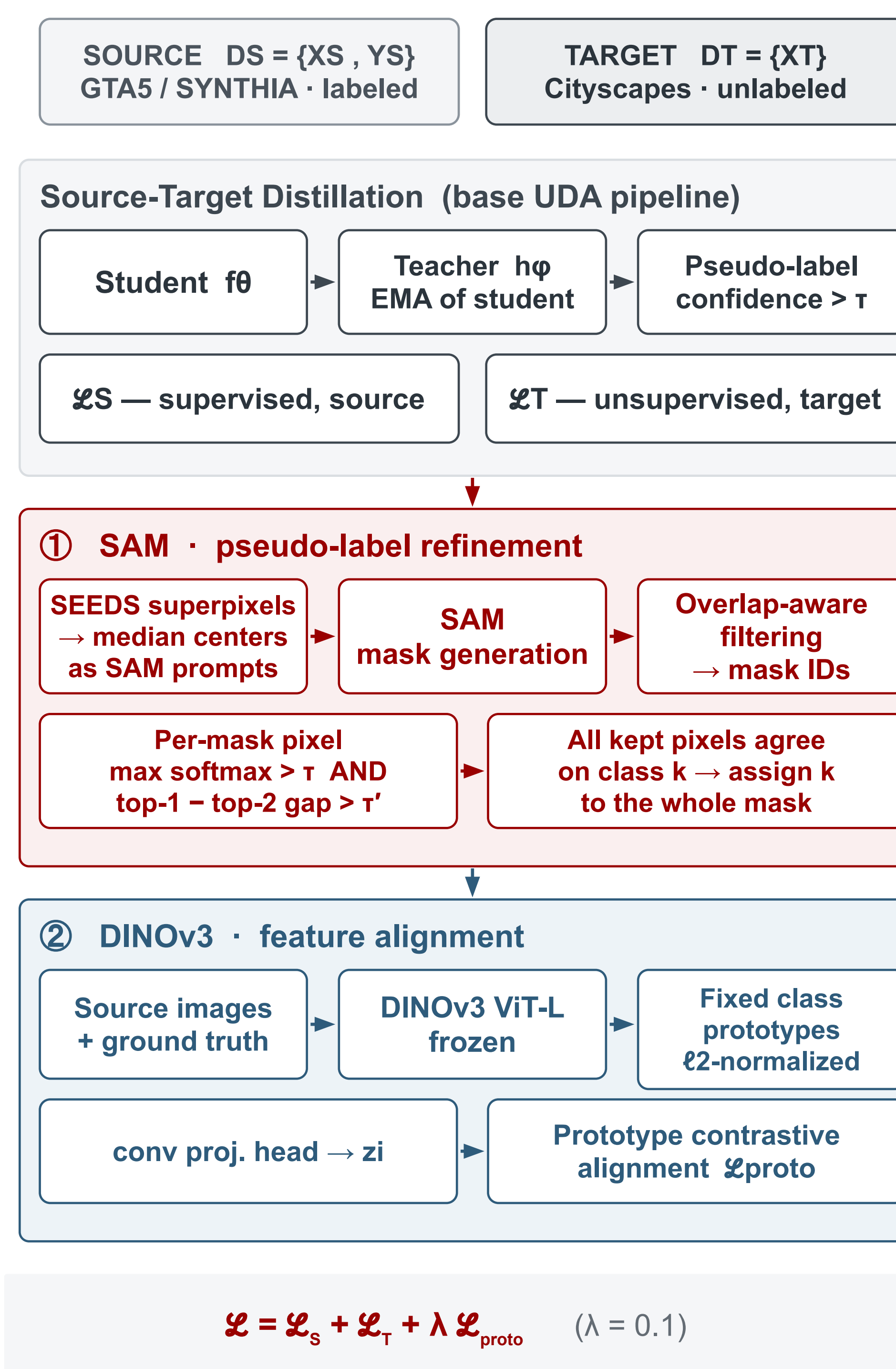
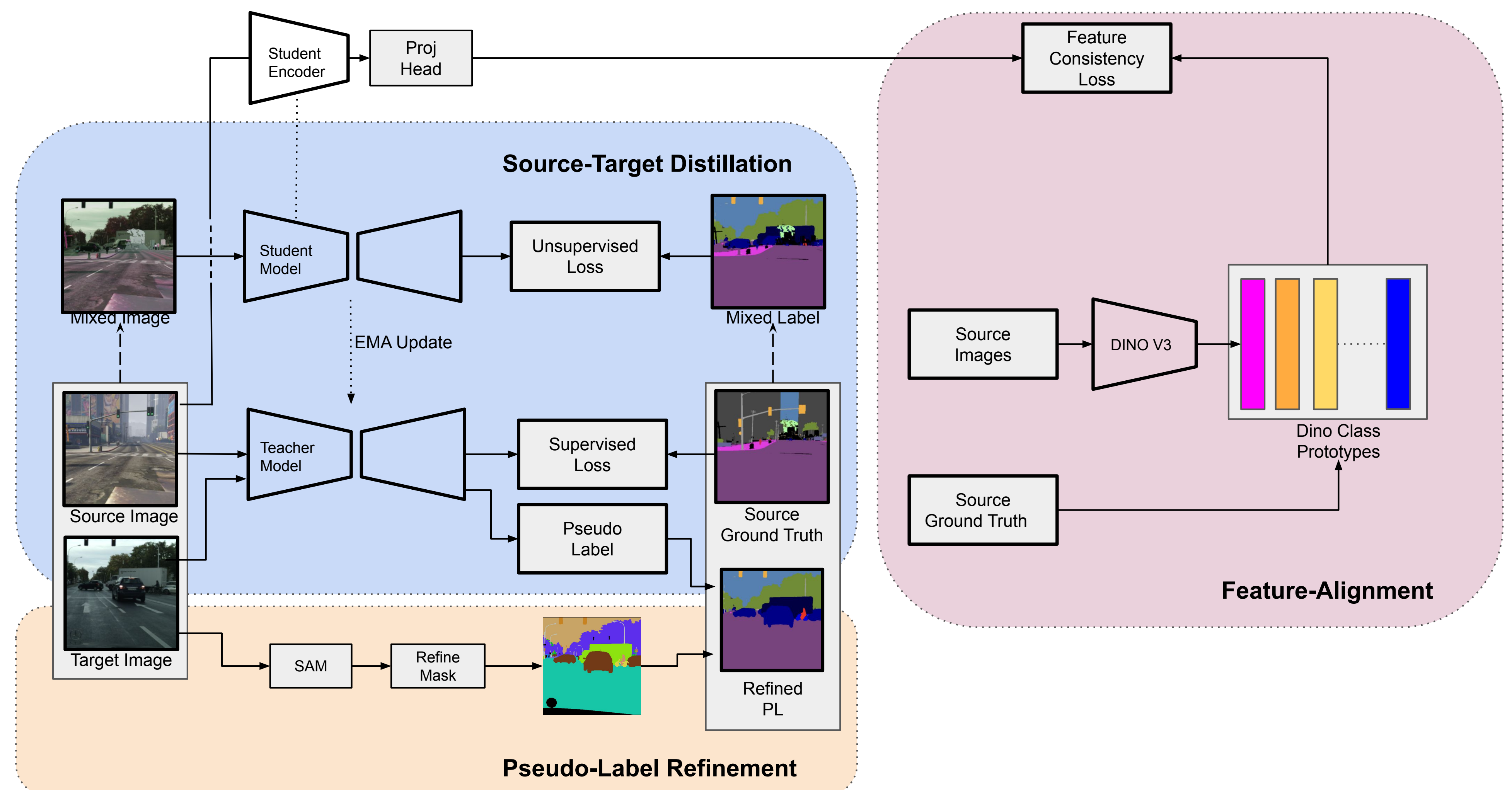


MIC [1]



Ours

03 Proposed Method



04 Results

GTA5 → Cityscapes · 19 classes

Method	mIoU (%)
DAFormer [2]	68.3
HRDA [3]	73.8
MIC [1] — base method	75.9
COPT [4]	76.1
Ours	77.4

SYNTHIA → Cityscapes · 16 classes

Method	mIoU (%)
DAFormer [2]	60.9
HRDA [3]	65.8
MIC [1] — base method	67.3
COPT [4]	67.4
Ours	68.8

+1.3 GTA5→CS **+1.4** SYNTHIA→CS

mIoU gain over COPT, the strongest (76.1→77.4 and 67.4→68.8).

05 Ablation

Backbone · UDA method	Base	+DINO	+SAM+DINO	Δ
DeepLabV2 · DACS	55.93	58.12	58.77	+2.84
DAFormer · DAFormer	66.72	68.45	69.17	+2.45
DAFormer · HRDA	73.72	74.57	76.04	+2.32
DAFormer · MIC	75.19	76.18	77.4	+2.21

Both components help on every backbone and framework tested. Prototype alignment helps DeepLabV2 most; SAM refinement pays off more as the baseline gets stronger.

07 References

- [1] Hoyer, L., Dai, D., Wang, H., Van Gool, L.: Mic: Masked image consistency for context-enhanced domain adaptation. In: CVPR (2023)
- [2] Hoyer, L., Dai, D., Van Gool, L.: Daformer: Improving network architectures and training strategies for domain-adaptive semantic segmentation. In: CVPR (2022)
- [3] Hoyer, L., Dai, D., Van Gool, L.: Hrda: Context-aware high-resolution domainadaptive semantic segmentation. In: ECCV (2022)
- [4] Mata, C., Ranasinghe, K., Ryoo, M.S.: Copr: Unsupervised domain adaptive segmentation using domain-agnostic text embeddings. In: ECCV (2024)